Flixbus Pricing Analysis – Explanation

1. Objective

The objective of this analysis is to identify cases where Flixbus pricing deviates from the market by comparing it with similar bus listings. The system flags instances where prices are significantly higher or lower than comparable options and quantifies the deviation.

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2. Defining Similar Buses

To ensure a fair comparison, buses are considered "similar" based on the following parameters:

• Date of Journey (DOJ)

Only buses operating on the same date are compared, as pricing varies significantly across different days due to demand fluctuations.

• A/C vs Non-A/C

Buses are segmented based on A/C availability since they belong to different pricing tiers and customer segments.

• Bus Type (Seater / Sleeper / Mixed)

Bus types are grouped into categories such as seater, sleeper, and seater-sleeper. Sleeper buses typically command a higher price, making it important to compare within the same category.

• Departure Time (±2 hours)

A time window of ±2 hours is used because customers generally consider buses within a nearby departure time range as substitutes.

• Journey Duration (±60 minutes)

Duration acts as a proxy for route length and service type. A tolerance of ±60 minutes ensures only comparable journeys are considered.

• Fallback Logic

If fewer than two comparable buses are found:

- The duration constraint is relaxed first
- If still insufficient, the entire group is considered

This ensures robustness even when data is sparse.

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3. Pricing Benchmark

• Weighted Average Price (WAP)

The system uses Weighted Average Price (WAP) to represent the overall price of a bus, as it accounts for different seat categories.

• Median as Benchmark

The median price of comparable buses is used instead of the mean because:

- It is less affected by outliers
- It provides a more stable and realistic benchmark

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4. Price Flagging Logic

The system compares Flixbus price against the median of comparable buses.

• TOO HIGH

Flagged when:
Flixbus Price > Median × 1.10

This indicates potential loss of demand due to higher pricing.

• TOO LOW

Flagged when:
Flixbus Price < Median × 0.90

This indicates possible revenue loss due to under-pricing.

• OK

If the price falls within ±10% of the median, it is considered acceptable.

• Threshold Justification

A ±10% threshold is used as a standard pricing tolerance in revenue management, balancing sensitivity and stability.

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5. Output Interpretation

For each Flixbus listing, the system provides:

- Number of comparable buses
- Peer price statistics (min, median, mean, max)
- Price difference (absolute and percentage)
- Flag status (TOO HIGH / TOO LOW / OK)
- Magnitude of deviation

This enables quick identification of pricing issues and prioritization of corrective actions.

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6. Assumptions

- Bus brand, service quality, and ratings are not considered
- Customers compare buses primarily based on time, type, and price
- Same-day comparison is sufficient for pricing decisions
- Availability of at least minimal comparable data is assumed

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7. Automation Plan (MVP)

The system can be automated using the following workflow:

1. Data Ingestion
   Fetch daily data from a data warehouse or API
2. Data Cleaning
   Standardize fields such as price, duration, and bus type using Python (pandas)
3. Peer Matching
   Group data based on similarity criteria and apply filtering logic
4. Price Comparison
   Calculate median prices and flag deviations
5. Output Generation
   Export results to Excel or a database
6. Alerts
   Send alerts via email or Slack for flagged cases
7. Scheduling
   Automate execution using cron jobs, AWS Lambda, or similar tools

Tools Used:

- Python (pandas, numpy)
- Excel (openpyxl)
- Optional: SQL, Power BI, cloud scheduling tools

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8. Use of AI Tools

AI tools such as ChatGPT/Claude were used for structuring the solution, debugging the script, and optimizing performance. All final logic, assumptions, and decisions were reviewed and validated independently.

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9. Key Insights (Optional)

- A portion of Flixbus listings are priced above market median, indicating potential loss of competitiveness
- Some listings are underpriced, suggesting opportunities for revenue optimization
- Pricing variation is higher in certain bus categories such as sleeper services
- Departure time plays a significant role in pricing differences

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10. Conclusion

This system provides a scalable and data-driven approach to monitor pricing performance. By identifying deviations from market benchmarks, it helps optimize pricing strategy, improve competitiveness, and maximize revenue.